

SEVAN, Armenia

WMO: 377090

Lat: 40.550N

Lon: 44.933E

Elev: 1937

StdP: 80.12

Time Zone: 4.00 (E04)

Period: 82-91

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-17.9	-15.3	-21.8	0.7	-16.5	-19.3	0.9	-14.3	17.4	-7.4	15.9	-7.5	2.1	290	0.671

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.2	24.0	14.6	22.7	14.3	21.3	13.9	16.8	21.3	15.8	20.2	15.1	19.4	4.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.1	13.6	19.5	14.2	12.8	18.4	13.4	12.1	17.7	55.2	21.3	52.0	20.3	49.4	19.4	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 14.1	10.1	8.7		-19.9	27.5	3.6	1.7	-22.5	28.7	-24.6	29.7	-26.7	30.6	-29.3	31.8	
(5)			DB	-20.3	19.1	3.5	2.6	-22.8	21.0	-24.9	22.5	-26.9	24.0	-29.4	25.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.0	-6.8	-6.9	-2.5	5.0	8.8	13.2	16.4	15.8	13.2	6.6	1.2	-4.7
	DBStd	9.10	3.77	4.07	4.81	3.30	2.89	2.65	2.29	2.49	3.05	3.64	3.96	4.83
	HDD10.0	2436	521	472	389	151	57	4	0	1	7	113	265	457
	HDD18.3	4881	779	706	647	401	295	154	66	84	157	363	514	715
	CDD10.0	611	0	0	0	1	19	100	199	179	102	9	0	0
	CDD18.3	14	0	0	0	0	0	1	7	5	2	0	0	0
	CDH23.3	75	0	0	0	0	0	2	29	29	14	0	0	0
	CDH26.7	3	0	0	0	0	0	0	2	1	1	0	0	0
(14) Wind	WSAvg	3.6	4.5	3.8	3.4	3.6	3.3	3.1	3.3	3.6	3.1	3.1	3.7	4.0
(15) Precipitation	PrecAvg	498	18	23	33	59	89	74	49	33	27	47	32	17
	PrecMax	703	52	58	72	116	173	165	114	93	98	85	82	52
	PrecMin	252	2	4	8	25	10	1	1	4	0	3	1	2
	PrecStd	98	13	13	16	23	39	33	30	21	25	23	21	11
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.5	3.2	9.0	15.6	19.4	23.1	25.7	26.0	25.1	18.4	12.6	6.6
		MCWB	-1.6	0.7	4.1	7.4	10.6	13.5	16.6	13.8	12.1	8.8	7.4	3.2
	2%	DB	0.2	1.6	6.3	13.6	17.5	21.2	24.0	23.8	22.6	15.9	9.5	4.6
		MCWB	-2.1	-0.4	2.2	6.8	9.9	13.2	15.8	14.3	11.8	8.0	4.7	1.9
	5%	DB	-1.0	0.5	4.4	12.1	15.7	19.6	22.6	22.4	20.7	14.0	7.8	3.3
		MCWB	-2.7	-1.3	1.4	6.3	9.4	12.8	15.5	14.4	11.4	7.6	4.2	0.8
	10%	DB	-2.0	-0.8	3.0	10.4	14.1	18.2	21.1	20.9	19.0	12.1	6.5	1.7
		MCWB	-3.4	-2.2	0.6	5.5	8.7	12.4	15.0	14.1	11.3	7.0	3.5	-0.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.8	1.2	4.5	8.7	12.3	15.4	19.2	17.5	15.1	10.2	7.5	3.6
		MCDB	0.4	2.8	7.8	13.2	16.7	19.7	23.5	22.2	19.5	15.6	12.4	5.5
	2%	WB	-1.6	0.0	2.7	7.7	11.0	14.5	17.4	16.3	14.0	9.1	5.7	2.3
		MCDB	-0.5	1.5	5.5	12.3	16.0	18.9	22.0	21.1	18.5	13.7	7.9	4.5
	5%	WB	-2.5	-1.1	1.7	6.8	10.0	13.7	16.4	15.5	13.0	8.3	4.7	0.9
		MCDB	-1.3	0.2	4.0	11.0	14.6	18.3	20.8	20.1	18.7	12.4	7.2	2.7
	10%	WB	-3.2	-2.0	0.8	5.9	9.2	12.8	15.6	14.7	12.1	7.6	3.8	0.0
		MCDB	-2.0	-0.7	2.7	9.7	13.1	17.4	19.8	19.1	17.7	11.3	6.2	1.4
(35) Mean Daily Temperature Range	5% DB	MDBR	5.6	6.6	6.8	8.0	8.2	8.4	8.2	8.8	10.6	8.8	6.2	5.5
		MCDBR	6.5	5.8	7.7	10.6	10.8	10.5	10.7	12.0	13.1	11.4	8.1	5.9
		MCWBR	5.1	4.5	4.9	5.3	5.2	5.2	5.3	5.1	5.2	5.5	5.1	4.1
	5% WB	MCDBR	5.5	5.2	7.0	9.4	9.9	9.4	9.0	9.5	11.4	9.7	7.5	4.9
		MCWBR	4.6	4.3	4.7	5.1	5.2	5.1	5.0	4.6	4.8	5.0	4.9	3.6
(40) Clear-Sky Solar Irradiance	taub		0.266	0.298	0.347	0.402	0.381	0.370	0.407	0.396	0.345	0.334	0.287	0.264
	taud		2.366	2.302	2.243	2.118	2.263	2.344	2.242	2.276	2.413	2.383	2.473	2.426
	Ebn at Noon		904	916	901	869	896	904	865	866	895	862	873	880
	Edh at Noon		83	104	124	151	134	124	136	128	104	95	75	73
(44) All-Sky Solar Radiation	RadAvg		2.15	3.07	3.65	4.43	5.47	6.47	6.43	5.97	4.82	3.21	2.21	1.70
	RadStd		0.24	0.45	0.35	0.55	0.31	0.44	0.40	0.46	0.34	0.27	0.30	0.24

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% DP	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (15 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page